

Chapter 16: Philosophy of AI

An Introduction to Universal Artificial Intelligence

Induction, Consciousness, Morality, Arguments For/Against AGI, Intelligence, Deep Learning

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Taylor & Francis / CRC Press, 2024

"Artificial intelligence would be the ultimate version of Google. The ultimate search engine that would understand everything on the web. It would understand exactly what you wanted, and it would give you the right thing."

— Larry Page

Chapter Overview

- 1 Introduction
- 2 16.1 Philosophy of Universal Induction
- 3 16.2 Consciousness, Free Will, and Other Qualia
- 4 16.3 Moral Considerations
- 5 16.4 Teleporting and Copying AGI
- 6 16.5 Arguments against AGI
- 7 16.6 Arguments for AGI
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Why Philosophy, at the End of a Technical Book?

Turing's seminal 1950 paper introduced, informally, many of the concepts this book has since made mathematically precise: what it means for a machine to be *universal* (Turing complete), and the idea of a machine self-modifying either by *screwdriver interference* (changing its own hardware) or *paper interference* (changing its own source code / information). While Turing's terminology is dated, the underlying philosophical questions are still very much alive:

- Should we expect AGI (Artificial General Intelligence) to arrive at all?
- Is it even something we *should* build?

This chapter revisits the philosophical foundations of everything developed so far (induction, agency, universal agents such as AIXI), and then turns outward to debate consciousness, morality, and the plausibility of AGI itself — closing with how modern Large Language Models (LLMs) relate to the Universal Artificial Intelligence (UAI) theory of this book.

Roadmap of the Chapter

- **16.1** revisits the philosophical backing (Epicurus, Occam, Bayes) of universal (Solomonoff) induction.
- **16.2–16.3** ask whether universal agents can be conscious, have free will, or deserve moral consideration.
- **16.4** asks whether AIXI-like agents would choose to replicate or teleport themselves.
- **16.5–16.6** weigh informal and formal arguments *against* and *for* the possibility of AGI.
- **16.7** formalizes *intelligence* itself via the Legg–Hutter (LH) measure, of which AIXI is the maximizer.
- **16.8** connects Deep Learning and LLMs back to Solomonoff induction and AIXI.
- **16.9** concludes the book.

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The Problem of Induction

Induction means drawing generalizations from data, or building a model of the world from observations, in order to *predict* the future from the past. The *induction problem* is an ancient philosophical puzzle (tracing back at least to Epicurus): no finite amount of past data can *logically guarantee* what will happen next, yet we clearly need some principled way to make such predictions.

Earlier chapters built a mathematical solution: **universal (Solomonoff) induction**. This performs Bayesian sequence prediction using a Bayes mixture starting from the **universal prior** $w_\nu^U = 2^{-K(\nu)}$ over the model class $\mathcal{M}_{sol}$ of all semicomputable semimeasures (ν is pronounced “nu” and denotes a candidate environment/hypothesis; $K(\nu)$ is the **Kolmogorov complexity** of ν – the length in bits of the shortest program that computes/describes ν). This section asks: *why* is this a good, philosophically sound choice?

Justification for the Model Class $\mathcal{M}_{sol}$

Why restrict to *computable* environments?

The choice of model class $\mathcal{M}_{sol}$ (all computable/semicomputable environments) is motivated by the **physical Church-Turing thesis** (Section 16.6.1): the assumption that all physical processes are computable.

- If the physical universe is (at least) computable, then by choosing the class of all computable environments we include *every* environment the agent could ever possibly encounter.
- A more pragmatic reason: if the true environment were *incomputable*, then by definition no computable agent could ever learn to predict it. There is nothing to be gained by worrying about environments no algorithm could ever handle.

Justification for the Prior: Three Principles in One Formula

The prior weight $w_\nu^U = 2^{-K(\nu)}$ combines three classical philosophical principles simultaneously:

- **Epicurus' principle:** if more than one theory is consistent with the observations, keep *all* of them. Since $2^{-K(\nu)} > 0$ for every ν , no environment is ruled out as impossible *a priori*, no matter how complex.
- **Occam's razor:** entities should not be multiplied beyond necessity — prefer the simplest theory consistent with the data. Kolmogorov complexity $K(\nu)$ gives an unbiased, formal notion of “simplicity”: a simpler environment has smaller $K(\nu)$, hence a *larger* prior weight $2^{-K(\nu)}$.
- **Bayes' rule:** the mathematical rule for updating a degree of belief given new evidence. The posterior $w^U(\nu | x_{<t})$ (belief in ν after observing data $x_{<t}$, i.e. all symbols before time t) shrinks toward zero if the data rules out ν , and grows if the data is consistent with ν .

Figure of merit

The number of prediction errors made using the universal prior is upper-bounded by a term proportional to $K(\mu)$, the complexity of the *true* environment μ (mu). No prediction scheme can do uniformly better, since that would imply a shorter description of μ than its own Kolmogorov complexity — a contradiction.

Table 16.1: Induction $\Leftrightarrow$ Deduction

Deductive reasoning derives the truth of a statement from axioms and logical rules with certainty; **inductive reasoning** instead assigns a probability in $[0, 1]$ representing our confidence. Remarkably, the two frameworks have a term-by-term correspondence:

	Induction	Deduction
Type of inference:	generalization/prediction	specialization/derivation
Framework:	probability axioms	logical axioms
Assumptions:	prior	non-logical axioms
Inference rule:	Bayes' rule	modus ponens
Results:	posterior	theorems
Universal scheme:	Solomonoff probability	Zermelo-Frankel set theory
Universal inference:	universal induction	universal theorem prover

Recent work has even tried to *combine* the two, e.g. assigning probabilities to quantified logical sentences, so that induction can be performed jointly with deduction rather than treating them as separate modes of reasoning.

Example 16.1.1: Induction for Brown and Blue Foxes

Setup. We repeatedly observe foxes and record their colour. Every fox seen so far has been brown. Two candidate hypotheses:

- H_1 : “All foxes are brown.”
- H_2 : “There exists a blue fox (somewhere, unseen).”

Key insight. Seeing more brown foxes increases our confidence in H_1 , but can *never fully confirm* it – a blue fox may always be hiding, unobserved. Symmetrically, we should never assign *exactly* zero probability to H_2 (Epicurus), regardless of how many brown foxes we have seen.

What if we then see one blue fox? One might think this immediately refutes H_1 and sets its probability to exactly zero. But we should always keep a degree of skepticism about any single observation (sensor noise, a trick of the light, a painted fox. . .). So seeing a blue fox provides very strong evidence against H_1 , driving its posterior probability very close to zero, but not *exactly* zero.

Python: A Toy Universal-Prior Induction Demo

Each hypothesis ν is weighted a priori by $2^{-\ell(\nu)}$ ($\ell(\nu)$ = description length, our stand-in for $K(\nu)$), updated via Bayes' rule after each observation.

```
1  # Two toy hypotheses, weighted a priori by 2^-ell(nu) (proxy for K(nu)):
2  #   H1 = "all foxes brown"                                ell(H1) = 3 bits
3  #   H2 = "brown, but a blue fox exists somewhere" ell(H2) = 8 bits
4  ell = {"H1_all_brown": 3, "H2_blue_exists": 8}
5  prior = {h: 2.0 ** (-1) for h, l in ell.items()}
6  Z = sum(prior.values())
7  prior = {h: p / Z for h, p in prior.items()}          # normalize
8  lik_brown = {"H1_all_brown": 1.0, "H2_blue_exists": 0.999} # P(brown|H)
9  lik_blue  = {"H1_all_brown": 1e-6, "H2_blue_exists": 0.001} # P(blue|H)
10
11 post = dict(prior)
12 for step, obs in enumerate(["brown"] * 5 + ["blue"], start=1):
13     lik = lik_brown if obs == "brown" else lik_blue
14     post = {h: post[h] * lik[h] for h in post}
15     Z = sum(post.values())
16     post = {h: p / Z for h, p in post.items()}          # Bayes' rule
17     print(f"#{step} ({obs}): P(H1)={post['H1_all_brown']:.6f} "
18           f"P(H2)={post['H2_blue_exists']:.6f}")
```

Output (abridged): $P(H_1)$ climbs from ≈ 0.90 to ≈ 0.9999 after 5 brown foxes, then *collapses* to ≈ 0.0009 after one blue fox — but never to exactly zero.

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Qualia: Consciousness, Free Will, Sentience

A recurring question in AI discussions is whether an AGI would possess or experience **consciousness**, **free will**, **emotions**, or **sentience** – collectively called *qualia* in philosophy – and whether this matters morally (could an AGI feel pleasure or pain the way humans do?). To make progress we must try to formally define these terms rather than debate them purely intuitively.

Free Will

The argument for: an intelligent agent has “choice” among a set of possible actions, so it seemingly has free will.

The counter-argument: the agent’s action at every step is fully determined by its policy π (π , a function from history to action) together with the history of events so far. Even a *stochastic* policy does not rescue free will, since the agent must still passively respect the outcome of an uncontrollable random process.

A reinforcement-learning agent, for instance, is confined to actions that (approximately) maximize its Q -value (the expected future value of taking a given action). It can at best choose freely *among ties* – among actions of equal Q -value. The book notes this “free will paradox” has a resolution developed elsewhere in the text (see the chapter on universal agents), but the tension between determinism and the intuitive feeling of choice remains philosophically deep.

Do RL Agents Feel Pain and Pleasure?

To even ask this question we must define “happy” / “sad” formally.

- **Naive definition:** an RL agent is happy when it receives high positive reward, sad/depressed when it receives low (very negative) reward.
- **Suicidal agents:** we saw earlier in the book that if reward is sufficiently negative for sufficiently long, a reward-maximizing agent may choose “suicide” (ending the episode) if that option is available. Is such an agent then “sad” or “depressed”?
- **Alternative definition:** happiness is not about the *absolute* reward level but about an *unexpected positive change* relative to the agent’s current best estimate – i.e. learning from the error between expected and actual reward. This is exactly **temporal difference (TD) learning**, a paradigm of reinforcement learning with a documented link to dopamine responses observed in primates in neuroscience studies.

Does AIXI Have Subjective Experiences?

This connects to the **hedonic treadmill**: human happiness tends to revert to a baseline after any major positive or negative life change.

Self-awareness. AIXI does not explicitly contain a model of itself: the hypothesis space AIXI considers is not broad enough to contain AIXI itself (any environment that includes a self-model of AIXI would not be computable, since AIXI's own decision process is uncomputable). However AIXI may still possess an *approximate* self-model and a form of implicit self-awareness, in the sense that it considers the effects of its own actions on the environment while planning, and maximizes reward with this in mind.

Pragmatic stance taken in the book: if consciousness, emotions, etc. are *necessary* for intelligent behaviour, we should expect them to emerge in agents like AIXI, since AIXI is by construction the most intelligent agent possible (with respect to the Legg–Hutter intelligence measure, Section 16.7). AIXI would display these properties *if and only if* they are useful for acting intelligently.

Formalizing Consciousness: Integrated Information Theory

One serious attempt to formalize consciousness is **Integrated Information Theory (IIT)**. Its axioms are:

- consciousness and experience exist;
- each experience is independent of external observers (intrinsic existence);
- consciousness is structured (composition);
- consciousness is specific: each experience is the particular way it is (information);
- consciousness is unified: each experience is irreducible to non-interdependent components (integration);
- consciousness is definite (exclusion).

From these axioms, IIT derives a measure of complexity Φ (capital phi) that is claimed to strongly correlate with consciousness.

A problem with IIT

Under this theory, certain very simple structures (e.g. expander graphs and 2D grids of XOR gates) are assigned implausibly *high* Φ , despite seeming unintelligent. This echoes the Chinese room argument (Section 16.5.2): a formal criterion can behave in counter-intuitive ways. IIT is at least a formal starting point; less formal proposals avoid this kind of falsification only by risking “not even being wrong.”

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Weak vs. Strong AI Hypothesis

Can machines *think* the way humans do, or do they merely *act as if* they think, without any internal experience? This gives rise to two hypotheses:

Definition: Weak vs. Strong AI

- 1 **Weak AI hypothesis:** an artificial intelligence can take actions *as if* it were intelligent.
- 2 **Strong AI hypothesis:** it is possible to build a machine that can *think* in the same sense a human can – it is self-aware and has a conscious experience of the world.

Most researchers take the **weak** AI hypothesis for granted, and on its face this seems true in many domains already. From a pragmatic perspective, it does not matter whether the strong hypothesis holds, so long as the agent performs the tasks required of it.

Moral Relevance of the Strong AI Hypothesis

From a *moral* perspective, however, the strong hypothesis matters a great deal. Comparing a modern autonomous vacuum cleaner to a human worker indentured to clean for no pay: the latter raises concerns of moral harm; the former, currently, does not.

We currently give little thought to the “welfare” of robotic vacuum cleaners, recommendation algorithms, or smart assistants (though some people do become emotionally attached to the latter). It is conceivable that a future super-intelligent agent may be ascribed moral status for the computations it performs.

Real-world echo: in June 2022, Blake Lemoine, a Google engineer, publicly raised concerns that *LaMDA*, a Large Language Model (LLM), might be sentient, based on his conversations with it.

The Neural Substitution Argument

One argument *in favour* of the strong AI hypothesis (though currently technologically out of reach):

"Suppose in the future, the process by which neurons operate are fully understood and characterized, to the point where synthetic neurons can be created to replace biological neurons, say, as a treatment for neurological diseases like Alzheimer's. A patient comes to the clinic for surgery and an awake craniotomy is performed, exposing the brain while the patient is still conscious. Nanobots are injected into the brain, where they navigate around, scanning for brain tissue at risk and replacing it with a synthetic drop-in replacement. The patient reports no loss or change in consciousness, feelings, other qualia, or memory. In extreme cases, the entire brain might slowly be replaced by a synthetic equivalent without the patient noticing any change in experience."

Physicalism vs. Dualism, and Brain Uploading

The assumption that consciousness/personality reside entirely in the neurological connections of the brain is called **physicalism**, as opposed to **dualism**, where mental processes cannot be explained solely as an emergent phenomenon of physical processes. If synthetic neurons perform the same function as biological ones, the patient should act and feel identical – it seems arbitrary to assign moral value before the procedure but not after.

Brain uploading / digital immortality: a related idea is hosting the pattern that makes up a brain on a server, simulating a virtual environment and piping in simulated sensory data. Even though everything is simulated, it would feel just as real to the “person” inside (cf. the *brain-in-a-vat* thought experiment). Such technology has obvious dangers in the wrong hands.

OpenWorm is a real, active open-source project aiming to simulate *Caenorhabditis elegans* (a roundworm) at the full cellular level – a small but concrete step in this direction.

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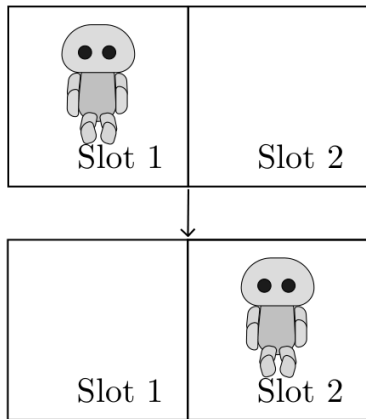
Would Intelligent Agents Replicate Themselves?

Digital agents can make copies of themselves cheaply and at high fidelity, so self-replication could plausibly be an action a sufficiently capable agent would take.

Within the AIXI framework: if self-replication leads to higher *expected reward*, then AIXI will replicate itself. This has been formally studied by extending AIXI to a setting where AIXI and AI_μ (the ξ -optimal and μ -optimal agents, respectively) have the option to make copies of themselves, or to “cut-and-paste” (effectively teleport) themselves, destroying the originals.

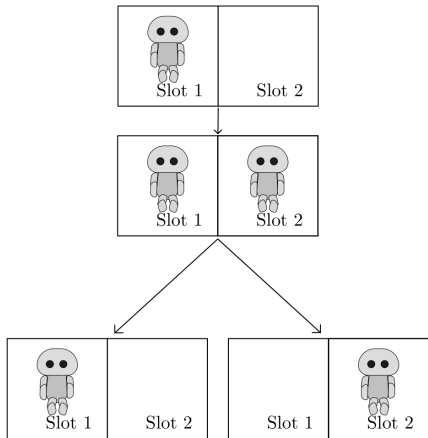
- A **copy-centered** agent's value function includes the observations from *all* its copies.
- A **static slot-centered** agent's value function considers only the observations that will come from a fixed, given slot number i .
- A **dynamic slot-centered** agent instead *estimates* which slot number it is in.

Figure 16.2: Cut-and-Paste Teleportation



First form of teleportation: the agent is cut-and-pasted into a new slot. This is essentially the same as *moving the position* of the agent – like a “wormhole” teleportation – where only the agent’s *location* changes, and there is exactly one continuous agent throughout (continuity of existence is preserved).

Figure 16.3: Copy-Paste-and-Delete Teleportation



Second form of teleportation: the agent is first *copied* into a new slot; then *one* of the two resulting copies is deleted. This mirrors the sci-fi trope (e.g. *Star Trek*) where an agent is scanned and reconstructed from local material at the target location, then the “original” is destroyed. Unlike cut-and-paste, there is momentarily *two* copies of the agent before one is deleted.

Who Chooses to Teleport?

- In **both** types of teleportation, a **copy-centered** agent will always choose to teleport, since it values having at least one copy of itself receive future observations, and teleporting does not reduce that (indeed cloning can only help).
- A **slot-centered** AI_μ agent will *not* teleport, since AI_μ acts to preserve exactly itself (its known slot), and teleporting risks losing its own identity/slot.
- A **slot-centered AIXI** agent, by contrast, *may* choose to teleport, precisely because it is uncertain about which slot it currently occupies – teleporting can be informative or advantageous under that uncertainty.

This mirrors an earlier open question in the book about whether AIXI or AI_μ would choose to *self-modify*: the answer depends entirely on how the value function is defined, not on some environment-independent notion of “identity.”

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Overview: Arguments against AGI

There is a long history of important thinkers arguing that computers will never do certain things at all, or as well as humans. Most of these arguments, the book argues, break down once they are made fully rigorous/formal. We cover four representative arguments:

- ❶ **Informal arguments of disability** (16.5.1): “a computer could never do X .”
- ❷ **Chinese room argument** (16.5.2): mimicking intelligence is not the same as *understanding* or being conscious.
- ❸ **No Free Lunch theorems** (16.5.3): averaged uniformly over all problems, all algorithms perform equally (poorly) – so universal solutions (hence AGI) might seem impossible.
- ❹ **Penrose–Lucas arguments** (16.5.4): a Gödel-like sentence that is true, but that no machine can ever prove is true.

16.5.1 Informal Arguments of Disability

The *argument of disability*: there exist tasks humans can do that an AI never can – play chess, translate languages, read faces, write poetry, compose music, hold a conversation, etc.

Historical pattern: a task once thought impossible for a computer is often eventually solved (or attempted with reasonable performance) once it is formalized, the right algorithmic approach is found, and/or enough compute/training data becomes available:

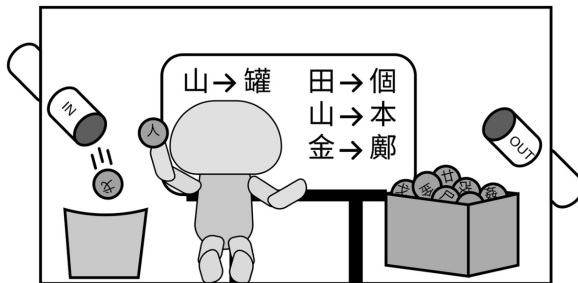
- Chess was thought unsolvable, until it was (Deep Blue, 1997).
- Go, with its enormous branching factor requiring deep intuition, was thought unsolvable, until AlphaGo (2016).
- Real-time strategy games with huge continuous action spaces and only pixel input (*Dota 2*, *Starcraft 2*) were thought unsolvable, until they too were mastered.

Moving goalposts: once a task is solved by AI, it is often no longer considered a “true” test of intelligence (e.g. chess-playing is now called mere “heuristic search” or “optimization”). AI-skeptics risk converging on the absurd claim that *no* task requires intelligence.

16.5.2 The Chinese Room Argument

Setup (Searle, 1980): imagine a room with a small input/output slot. Questions written in Chinese are inserted; answers (also in Chinese) are retrieved. Inside is a person who speaks *no* Chinese, but has an instruction book (in English) telling them exactly how to match input symbols to output symbols.

Figure 16.4.



From the outside, the room appears fluent in Chinese. But the person inside does not understand a word – they are mechanically following instructions. The argument: an AI, similarly, may just be following a “book of instructions” without any real understanding.

Chinese Room: A First Counter-Argument

This does not refute the weak AI hypothesis (Section 16.3), which is the primary hypothesis this book is concerned with: our intelligence measure (Section 16.7) is based only on how the agent *acts*, not on the internal thought process, nor on whether it is conscious.

If we do care about the strong AI hypothesis, one may argue that in the Chinese room setup, the *process of using the book* could itself be ascribed intelligence, since it possesses perfect knowledge of the meaning of words, characters, and sentences in Chinese – and therefore, on some level, “understands” those meanings.

Chinese Room: Further Refutations

Objecting to the book itself. A book that lets someone converse fluently without *any* understanding could not be a simple phrase-book or dictionary. It would need to break input sentences into words, look each up, construct a response, and translate back – so the user would need at least *some* understanding of Chinese structure. If the user truly had *zero* understanding, the book would need to be an astronomically massive lookup table, with a mechanism to search it quickly – and given the exponential (or infinite) number of possible sentences, such a book may not even fit in the accessible universe. We may reasonably ascribe understanding to whatever process *created* such a book.

The neuron analogy. One could argue human “understanding” is merely an emergent phenomenon of neurons firing – a single neuron clearly does not understand anything, so (by the same logic) no part of the brain understands, so humans do not either. This is absurd from our own perspective. **The fallacy:** assuming a whole cannot have a property that none of its parts have individually. No single bolt, nut, or rod in a car engine can transport people; only the assembled whole can.

16.5.3 No Free Lunch Theorems

The **No Free Lunch (NFL)** theorems state that, averaged uniformly over *all* possible problems (in a certain class), every algorithm performs equally – and equally poorly. This is popular among machine learners as a justification for domain-specific approaches, and has been used to argue against AGI (the ultimate domain-*agnostic* solver).

NFL is formally a version of what Hume called the *principle of uniformity of nature*, which (taken to the extreme) would invalidate all inductive reasoning. The theorems are mathematically correct, provided their key assumption holds: complete uniformity over problems.

Thesis 16.5.1 (Universal = M vs. uniform = NFL sampling)

- The universal distribution M is a *non-dogmatic* quantification of Occam's razor – a cornerstone of science and a solution to the induction problem.
- The uniform distribution over data is a *dogmatic* prior: the belief that the world is pure white noise.

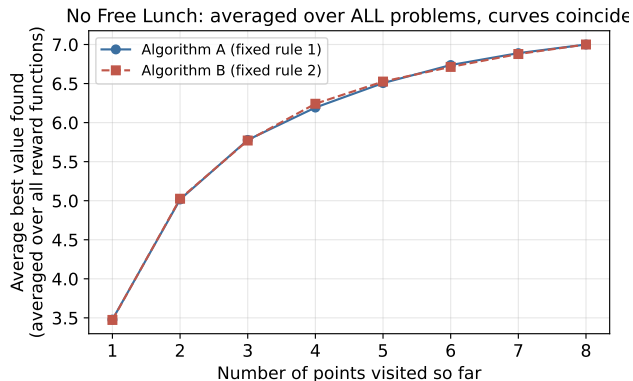
The fallacy: equating “uniformly distributed problems” with “uniformly distributed *data*.” The latter means data is (almost surely) white noise, on which induction *must* fail – but almost nobody cares about performing well on pure noise. If instead *problems* (programs) are uniformly distributed, induction via a universal Turing machine works extremely well, since output strings then have a built-in bias toward simplicity.

Python: No Free Lunch, Empirically

Averaged over *all* possible reward assignments on a tiny search space, two very different fixed search algorithms achieve identical average performance – illustrating the NFL theorem directly.

```
1 import numpy as np
2 rng = np.random.default_rng(0)
3 n_points, n_functions = 8, 3000
4 idx = np.arange(n_points)
5 def best_seen(order, values):
6     seen, out = -np.inf, []
7     for i in order:
8         seen = max(seen, values[i]); out.append(seen)
9     return out
10 avgA = np.zeros(n_points); avgB = np.zeros(n_points)
11 for _ in range(n_functions):
12     values = rng.permutation(n_points).astype(float) # random problem
13     orderA = rng.permutation(n_points) # Algorithm A: arbitrary traversal
14     orderB = idx.copy() # Algorithm B: always 0,1,2,...
15     avgA += best_seen(orderA, values)
16     avgB += best_seen(orderB, values)
17 avgA /= n_functions; avgB /= n_functions
18 print("Algorithm A:", np.round(avgA, 3))
19 print("Algorithm B:", np.round(avgB, 3))
20 # -> the two rows are (statistically) identical: NFL in action.
```

No Free Lunch: The Resulting Curves Coincide



Averaged over *all* possible reward functions, Algorithm A (a fixed “arbitrary-looking” traversal) and Algorithm B (the fixed cyclic traversal $0, 1, 2, \dots$) find, on average, exactly the same best value after visiting the same number of points – neither algorithm is better than the other once every problem is equally likely.

16.5.4 The Penrose–Lucas Arguments

The **Penrose–Lucas** argument is effectively an instance of the argument from disability (16.5.1), but formal and worth engaging with. It claims there exist true logical statements that an AI can never prove, but that humans can see are true *by reason* – a consequence of Gödel's incompleteness theorem.

Penrose: *“The inescapable conclusion seems to be: Mathematicians are not using a knowably sound calculation procedure in order to ascertain mathematical truth. We deduce that mathematical understanding – the means whereby mathematicians arrive at their conclusions with respect to mathematical truth – cannot be reduced to blind calculation!”*

Lucas: *“Given any machine which is consistent and capable of doing simple arithmetic, there is a formula it is incapable of producing as being true but which we can see to be true.”*

The Gödel Sentence for AIXI – and Its Mirror Image

The claimed Gödel sentence:

“AIXI cannot prove that this sentence is true.”

Penrose claims humans see this sentence is obviously true, while AIXI cannot prove it.

But we can construct the mirror-image sentence just as easily:

“Penrose cannot prove that this sentence is true.”

Now humans who are *not* Penrose, as well as AI systems, can see *this* sentence is true (by contradiction: if Penrose could prove it true, it would be false). Yet Penrose himself can never prove it.

Conclusion: although Penrose–Lucas raises an interesting point about the logical limits of *any* sufficiently powerful formal reasoning system, it is likely irrelevant to AGI: an inability to prove every true statement is far too strict a requirement for a definition of intelligence – human mathematicians also have limitations on recognizing unprovable truths, and make mistakes. Moreover, to even understand the self-referential sentence purely arithmetically, one would need to know AIXI’s exact source code, and AIXI would need to know its own soundness – something a suitably idealized “mechanical knower” provably cannot do for itself.

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Overview: Arguments for AGI

In contrast to Section 16.5, we now present philosophical and *experimental* arguments *for* the eventual possibility (or even near-term arrival) of AGI:

- ① **Physical Church-Turing thesis** (16.6.1): intelligent hardware (human brains) already exists, so replicating it in machines is an *engineering*, not a philosophical, problem.
- ② **Moore's Law** (16.6.2): computing power has doubled roughly every 18 months for a century, and has already surpassed naive estimates of raw human-brain compute.
- ③ **AI Progress** (16.6.3): the accelerating, broad track record of concrete AI successes across games, vision, language, and science.

16.6.1 The Physical Church-Turing Thesis

Physical Church-Turing Thesis (PCT / Church-Turing-Deutsch principle)

All physical processes are computable.

If **physicalism** is true (the behaviour of any object is defined entirely by its physical properties), then PCT applied to the human mind implies that all behaviour of the human mind is computable – ruling out something incorporeal (a “soul”) being necessary for consciousness to work.

Since the human mind is an example of a physical process, under this assumption it is computable, so there *exists* a Turing machine that could (in principle) emulate it: scan a human brain, and simulate the scan on a computer. The brain fires at only around 200 Hz, yet performs complex tasks – suggesting it is a slow but highly parallel device, so a faithful simulation could likely run much faster than a biological brain, appearing *super-humanly* intelligent from the outside (e.g. effectively getting 10 years to think between each sentence of a conversation).

Variations of the Physical Church-Turing Thesis

- **Extended Church-Turing thesis (ECT):** “A probabilistic Turing machine can *efficiently* simulate any realistic model of computation” (with ‘efficiently’ usually meaning within polynomial time).
- **Quantum complexity-theoretic Church-Turing thesis:** the quantum analogue – “A quantum Turing machine can efficiently simulate any realistic model of computation.”

Even if PCT (or its variants) turned out to be *wrong*, and incomputable physical processes were somehow essential for cognition, it would be implausible for such processes to occur *only* in human brains – it is more plausible we could build machines harnessing the same physics. If the brain turns out to be an efficient quantum computer, we could still build AGI on quantum computers – this would only *delay* AGI until the relevant technology matures, not rule it out. A caveat: quantum computers are not strictly *more powerful* than classical computers (they can be simulated classically); they may only be more *efficient* for certain problems (e.g. factoring).

16.6.2 Moore's Law

Moore's Law: computing power roughly doubles every 18 months (a faster pace than Moore's original yearly estimate), and this trend has held up remarkably well since inception.

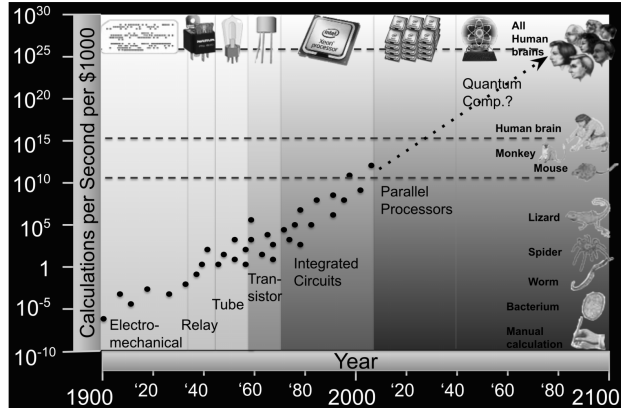


Figure 16.5: computing power (calculations per second per \$1000) vs. year, log scale, vs. estimated compute of biological brains.

Moore's Law and the Path to AGI

Global computing power has grown exponentially to the point of already exceeding estimates of raw human-brain compute (roughly 10^{17} logical operations per second). Assuming PCT, it would already be *theoretically* possible to simulate a human mind computationally – but having enough raw operations-per-second is not sufficient:

- little is known about *how* the brain actually operates at sufficient fidelity to simulate;
- the technology to *scan* a brain at sufficient resolution does not yet exist;
- the computational overhead of simulation may make brain simulation intractable for a few more decades of Moore's law, even if possible in principle.

It was predicted in 2016 that the classical doubling trend would slow around 2020–2025, but the (possible) rise of accessible quantum computers could provide a further computational leap analogous to the classical one.

16.6.3 AI Progress: Perception and Language

The single strongest piece of evidence for the eventual possibility of AGI is arguably the breadth of success AI has already achieved, especially since the deep learning era:

- **Computer vision:** extracting information from images; convolutional neural networks (CNNs) exceed human performance on many subproblems; *ResNet* uses skip connections to train very deep networks; *NeRF* represents 3D scenes with a neural network mapping position/viewing-angle to colour.
- **Speech recognition:** transcribing speech to text – now an end-to-end deep-learning task, nearly (but not fully) solved.
- **Machine translation & NLP:** from the 1960s chatbot *Eliza*, to *IBM Watson* winning *Jeopardy!* against human champions in 2011, to modern Transformer-based systems that can write articles indistinguishable from human writing.

AI Progress: Science, LLMs, and Multimodal Models

- **AlphaFold** (DeepMind): predicts 3D protein structure from amino-acid sequence with accuracy comparable to costly experimental methods – a landmark for AI-assisted science.
- **GPT-4 Turbo** (OpenAI, Nov. 2023): a decoder-only Transformer, scoring 90th percentile on the Bar Exam, and processing both visual and textual input. The key architectural ingredient is the **attention mechanism**, letting the model weigh the relative importance of every other position in a sequence, and allowing parallel (rather than strictly sequential) processing during training.
- **CLIP** (Contrastive Language-Image Pre-Training): jointly trains a vision Transformer and a text Transformer so that matching (image, caption) pairs map to nearby embedding vectors – producing a semantically meaningful representation used as input to many other models.

Generative AI and Diffusion Models

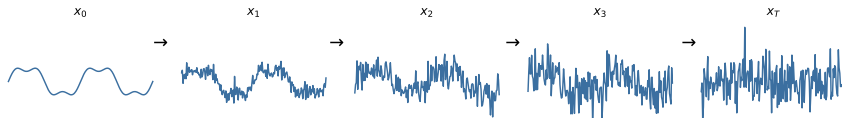
Diffusion models learn a complex data distribution (e.g. over images of animals) by defining a **diffusion process**: starting from a real datapoint $x_1 = x$, each successive term $x_{t+1} = x_t + \varepsilon_t$ adds a small amount of Gaussian noise ε_t (varepsilon, a random perturbation) according to a schedule, so that after T steps x_T looks like pure random noise (analogous to a drop of dye diffusing in water, driven by entropy).

Generation: the model learns to *undo* this process – de-noise x_{t+1} back towards x_t – so that starting from pure random noise and repeatedly applying the learned de-noising step produces a brand-new, realistic sample. *Latent* diffusion models first compress images into a smaller, semantically meaningful feature space (using e.g. a U-net and CLIP), making generation far more efficient. Applications include image generation/upscaling, video, music, and 3D mesh generation.

Python: A Numeric Forward-Diffusion Demo

```
1 import numpy as np
2 rng = np.random.default_rng(1)
3 n, T = 200, 5
4 t_axis = np.linspace(0, 4*np.pi, n)
5 x0 = np.sin(t_axis) + 0.3*np.sin(3*t_axis)           # a clean "signal"
6 alphas = np.linspace(0, 1, T) ** 1.6                # noise-fraction schedule
7 for step, a in enumerate(alphas):
8     xt = np.sqrt(1 - a) * x0 + np.sqrt(a) * rng.normal(size=n)
9     sig_var = np.var(np.sqrt(1 - a) * x0)
10    noise_var = np.var(np.sqrt(a) * rng.normal(size=n))
11    print(f"step {step}: alpha={a:.2f} sig-var={sig_var:.3f} noise-var={noise_var:.3f}")
12 # As alpha: 0 -> 1, sig-var shrinks to 0, noise-var grows to 1:
13 # x0 (clean sine wave) morphs smoothly into x_T (pure Gaussian noise).
```

Forward diffusion: $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$ (signal $\rightarrow$ noise)



AI Progress: Games as a Proxy for Intelligence

Games are popular because the goal, action space, and dynamics are well-defined, with communities of high-level human players to compare against. Notable milestones:

- **Checkers** (1959): early reinforcement learning beats its creator.
- **TD-gammon** (1995): self-play + TD learning reaches master-level Backgammon.
- **Deep Blue** (1997): defeats world chess champion Garry Kasparov using custom hardware and a hand-tuned evaluation function.
- **DQN** (2015): masters a large collection of Atari games at human level.
- **AlphaGo** (2016): Monte-Carlo tree search + deep neural networks defeat world Go champion Lee Sedol.

AI Progress: Games (continued) I

- **AlphaStar** (2019): combines supervised learning and RL to rank in the top 0.2% of *StarCraft* players.
- **Pluribus** (2019): a superhuman poker agent, using Monte Carlo counterfactual regret minimization plus self-play, beating elite professional poker players.
- **MuZero** (2020): learns *Go*, chess, *Shogi*, and Atari games using Monte Carlo Tree Search and self-play, with *no* prior domain knowledge – exceeding AlphaGo.
- **Agent57** (2020): a deep RL agent achieving human-level performance across *all* 57 games of the Arcade Learning Environment (ALE) benchmark.
- **Cicero** (2022): an LLM tuned with RL, ranked in the top 10% of online *Diplomacy* players – a game requiring tactical decision-making *and* negotiation with other (human) players.
- **Gato** (2022): a single multimodal agent excelling at a large collection of games, robotics, vision, and language tasks simultaneously.

The Role of Language in Intelligence

Language models such as GPT-3 and GPT-4 have been shown to generalize well beyond pure language tasks – playing chess, writing code, and passing theory-of-mind tests. Language can act as an efficient compression or coding scheme for knowledge, but there remain concepts that are fundamentally hard to convey *only* through language (e.g. a detailed image) – as the saying goes, “a picture is worth a thousand words.”

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What Is Intelligence?

Researchers have struggled for decades to define intelligence precisely. Common informal themes: the ability to solve a *wide class* of problems, display creativity/reasoning/deduction, and acquire and process *new* knowledge.

Crucially, we want a system that is good at *all* tasks (including ones it has never seen before), rather than a collection of narrow, built-in skills – this is the notion of **general** intelligence this chapter tries to formalize.

16.7.1 Legg–Hutter (LH) Intelligence: Informal

Synthesizing many definitions from psychology, philosophy, computer science, and neuroscience, Legg and Hutter proposed:

Legg–Hutter (LH) Intelligence (informal)

Intelligence measures an agent's ability to achieve goals in a wide range of environments.

This captures performing well across a *wide class* of problems, rather than excelling at one narrow task: a chess bot may play chess superhumanly, but cannot fill out tax forms, load a dishwasher, or compose poetry. It also excludes a “wise hermit” intelligence that thinks all day but takes no action affecting its environment – such an agent would be indistinguishable from an unintelligent one and serve no useful purpose. Requiring the agent to *achieve goals* captures this, while remaining agnostic to the strong-AI hypothesis (Section 16.3): only the agent's actions matter, not its internal state.

16.7.2 Other Intelligence Tests and Measures (I)

- **Raven test:** recognize a pattern in a sequence of geometric images and choose the missing piece. Narrow (pattern recognition), but many problems can be reframed as sequence prediction, so it is not entirely disconnected from general intelligence.
- **Turing test:** an interrogator chats (text only) with both a human and an AI; the AI “passes” if the interrogator cannot reliably tell them apart. Too strict: it tests mimicry of *human idiosyncrasy* (even including deliberately misspelled words) rather than intelligence – a genuinely intelligent alien species would likely *fail* it, not because it is unintelligent, but because it lacks familiarity with human culture and language. It is also purely binary (pass/fail), not a graded score.
- **Loebner prize** (1991–2019): a cash prize for the most human-like chatbot (judged by a human panel); the top \$100,000 prize (for fooling a judge completely) was never claimed.

16.7.2 Other Intelligence Tests and Measures (II)

- **Hutter prize:** a cash prize for how far enwik9 (a 1GB sample of Wikipedia) can be losslessly *compressed*. Motivation: compression is closely linked to intelligence – an “intelligent” compressor finds and exploits patterns/regularities in the data, using them to predict (and hence compress) further.
- **Chollet’s measure:** views intelligence as the ability to *adapt and generalize* to novel situations by recognizing patterns and re-using previously learned knowledge – emphasizing transfer of knowledge across domains, in contrast to traditional IQ tests or task-specific performance measures.
- **AI-Complete** (informal, by analogy with NP-Complete): problems for which *any* algorithm solving them must be generally intelligent – e.g. fluent human-level translation, automatic peer review of research papers, or proving prize-worthy mathematical theorems.

16.7.3 Formalizing the LH-Intelligence Measure: Design Choices

We want to define general intelligence as a *weighted average* of how well an agent achieves “the goal” across a large class of environments. This raises several design questions:

- **Goals:** too simple a goal makes every agent “intelligent by definition”; too internal a goal lets the agent cheat by picking an easy target. Solution: the goal is communicated to the agent *by the environment*, via rewards, within the standard cybernetic agent-environment model – the goal is always maximizing the sum of rewards (the *reward hypothesis*).
- **Balance:** should easy environments (where any agent does well) dominate the average? No – we want to penalize agents heavily for failing simple environments, but only lightly for failing difficult ones (since few agents can succeed there at all).

16.7.3 Formalizing LH-Intelligence: Universality and Balance

Universality (avoiding a discount-rate parameter). Optimal behaviour depends on how far-sighted the agent is (its discount factor γ , gamma). To avoid making intelligence dependent on this arbitrary choice, and to prevent some environments dominating just by offering disproportionately large rewards, we restrict to **reward-summable** environments: those where the total reward received can never exceed 1.

Bounded value function

$$V_{\rho}^{\pi} := \mathbb{E}_{\rho}^{\pi} \left[\sum_{i=1}^{\infty} r_i \right] \leq 1$$

for any policy π and environment ρ (rho). $\mathbb{E}_{\rho}^{\pi}$ is the expectation of the total reward $\sum r_i$ collected when policy π acts in environment ρ ; bounding it by 1 removes the need for discounting altogether – how quickly an environment “spends” its limited reward budget determines whether short- or long-term planning is required, rather than an externally imposed discount rate.

16.7.3 Formalizing LH-Intelligence: Environment Class and Weighting

Environment class $\mathcal{M}$: chosen to be as general as possible – *any* computable stochastic environment. Incomputable environments are excluded, since there would be no way to even accurately predict what percepts/rewards they would return (and the physical Church-Turing thesis, Section 16.6.1, makes considering them unnecessary anyway).

Weighting: not all environments are equally difficult. We use Kolmogorov complexity K as a proxy for difficulty: environments with a short description (in the Kolmogorov sense) are expected to be easier to perform well in, since the agent has fewer bits to learn to identify the true environment. Hence the natural weighting is the **universal prior** $2^{-K(\mu)}$ over environments μ – the very same prior used for universal induction in Section 16.1. This choice makes the intelligence measure itself an embodiment of Occam's razor.

16.7.4 Legg–Hutter (LH) Intelligence: Formal Definition

Definition 16.7.1 (Legg–Hutter intelligence)

The intelligence $\Upsilon(\pi)$ (capital upsilon) of a policy π is the undiscounted value function of π , averaged over all reward-summable computable stochastic environments $\mathcal{M}$, weighted by the Kolmogorov complexity of each environment:

$$\Upsilon(\pi) := \sum_{\nu \in \mathcal{M}} 2^{-K(\nu)} V_{\nu}^{\pi}(\epsilon)$$

where $V_{\nu}^{\pi}(\epsilon) = \mathbb{E}_{\nu}^{\pi}[\sum_{t=1}^{\infty} r_t]$ is the expected total reward earned by π in environment ν , starting from the empty history ϵ .

In words: intelligence is the weighted average of how well π performs across every computable, reward-summable environment ν , where simpler environments (lower $K(\nu)$) count for more.

Theorem: AIXI Maximizes LH Intelligence

Theorem 16.7.2 (AIXI maximizes LH intelligence)

$$\pi^{\text{AIXI}} := \pi_{\xi}^* = \arg \max_{\pi} \Upsilon(\pi), \quad \bar{\Upsilon} := \max_{\pi} \Upsilon(\pi) = \Upsilon(\pi^{\text{AIXI}})$$

Here $\pi^{\text{AIXI}} = \pi_{\xi}^*$ is the (undiscounted) Bayes-optimal policy that maximizes expected value under the universal mixture ξ (x_i); $\bar{\Upsilon}$ denotes the highest LH-intelligence score any policy can achieve.

AIXI is precisely this Bayes-optimal agent from earlier chapters, and it achieves the *maximum possible* LH-intelligence score $\bar{\Upsilon}$. The gap $\bar{\Upsilon} - \Upsilon(\pi)$ measures exactly how far any other agent π falls short of the theoretical optimum.

Why This Definition Is a Good One

The Legg–Hutter definition is claimed to be:

- **valid** (well-motivated by the informal definition),
- **meaningful** (high-scoring agents do well in most simple and many complex environments),
- **informative** (gives a graded numerical score, not pass/fail),
- **wide-range** (orders everything from random policies to super-intelligent AIXI),
- **general** (covers the entire class of computable environments, the largest class we could ever need),
- **unbiased** towards culture/language (unlike the Turing test), **fundamental** (built from computability, complexity, and information theory), and **objective (non-anthropocentric)**: unlike most tests, it does not measure relative to how *humans* behave – a super-intelligent agent with zero cultural knowledge would fail the Turing test yet still score very highly here).

Drawbacks of LH Intelligence

- It is **incomputable**, since it depends on Kolmogorov complexity K , which is itself incomputable. Approximations are needed in practice (Section 16.7.5).
- There is a hidden parameter: the choice of the reference **universal Turing machine** U used to define K . There is no known “canonical” choice, and this has been an open problem in algorithmic information theory since its inception – some choices of U can even lead to nonsensical intelligence rankings.
- One could weaken the environment class to something computable, like (partially observable) Markov decision processes, replacing the incomputable weight $2^{-K(\mu)}$ with a computable substitute.

16.7.5 Approximations of Intelligence: AIQ

Practical psychometric tests (Stanford-Binet, Wechsler, Raven, etc.) are not *general enough* to be true intelligence tests – they capture memory, reasoning, pattern-recognition, etc. separately.

Artificial Intelligence Quotient (AIQ) approximates $\Upsilon(\pi)$ via *sampling*: environments/programs p are sampled according to $2^{-\ell(p)}$ ($\ell(p)$ = program length), which approximates the simplicity weighting $2^{-K(\mu)}$ (since $K(p) \approx \ell(p)$ for most short programs p). The AIQ estimator is:

AIQ estimator

$$\hat{\Upsilon}(\pi) := \frac{1}{N} \sum_{i=1}^N \hat{V}_{p_i}^{\pi}$$

where $\hat{V}_{p_i}^{\pi} = \sum_t r_t$ is the empirical return of π on the i -th sampled program p_i , and N is the number of sampled programs. On average, $\hat{V}_{p_i}^{\pi}$ equals the true expected return $\mathbb{E}_{p_i}^{\pi}[\sum_t r_t]$.

A **BrainFuck (BF)** interpreter is used as the reference machine: an esoteric but Turing-complete language with only 8 symbols, chosen because it is simple, unbiased (no built-in domain knowledge), and produces very short programs for simple environments.

Python: A Toy LH-Intelligence / AIQ Worked Example

```
1 import numpy as np
2
3 # --- (a) Toy LH-intelligence: weighted sum over 5 environments ---
4 weights = np.array([0.50, 0.25, 0.125, 0.0625, 0.0625]) #  $\sim 2^{-K(nu)}$ ,  $nu=1..5$ 
5 weights /= weights.sum()
6 general_agent = np.array([0.90, 0.85, 0.75, 0.65, 0.55]) #  $V^{pi\_nu}$  for  $pi_1$ 
7 narrow_agent  = np.array([0.05, 0.05, 0.05, 0.05, 0.95]) #  $V^{pi\_nu}$  for  $pi_2$ 
8
9 epsilon_general = float(np.sum(weights * general_agent))
10 epsilon_narrow  = float(np.sum(weights * narrow_agent))
11 print(f"Upsilon(general) = {epsilon_general:.3f}") # 0.828
12 print(f"Upsilon(narrow)   = {epsilon_narrow:.3f}")  # 0.106
13
14 # --- (b) Toy AIQ: sampling-based estimator, converging as N grows ---
15 rng = np.random.default_rng(42)
16 N = 4000
17 lengths = rng.geometric(p=0.15, size=N) # program lengths  $\sim 2^{-l(p)}$ 
18 true_mean = 1.0 / (1.0 + 0.15 * lengths) # simpler => higher value
19 values = np.clip(true_mean + rng.normal(scale=0.15, size=N), 0, 1)
20 running_avg = np.cumsum(values) / np.arange(1, N + 1)
21 print(f"AIQ estimate after N={N}: {running_avg[-1]:.3f}")
```

Takeaway: the general agent has far higher LH-intelligence Υ despite *never* being the best in any single environment – breadth, weighted by simplicity, is what the measure rewards.

Atari (ALE) as a Practical Intelligence Measure

The **Arcade Learning Environment (ALE)**, a large collection of Atari 2600 games, is a popular practical intelligence proxy: doing well across *many* different games is a weak form of general intelligence. Deep Q-Networks (DQN) were the first major success; later agents such as Agent-57 outperformed humans on all 57 ALE games.

A handful of games (*Skiing*, *Private Eye*, *Pitfall*, and especially *Montezuma's Revenge*) remain very hard for most agents, because they combine three notoriously difficult RL problems:

- **Exploration-exploitation trade-off**: balancing learning more about the environment vs. exploiting what is already known.
- **Long-term credit-assignment problem**: rewards only arrive long after the actions that caused them.
- **Partial observability**: the current percept does not contain everything needed to determine what happens next.

Interestingly, agent performance on a carefully chosen subset of just 5 Atari games has been shown to be already indicative of average performance across *all* ALE games.

Solving One Very General Problem

Throughout the chapter, “intelligence” has meant good performance across a *wide class* of (typically simpler) environments. An alternative is to consider one, single, very *general* environment requiring a large skill set – e.g. a self-driving car delivering passengers, which requires computer vision, path-finding, object detection/avoidance, and trade-offs between comfort, safety, and speed.

Interestingly, an agent that does well *only* in such a broad, single environment, but poorly elsewhere, will still have **low** LH-intelligence Υ : a single very general environment μ_{general} still only gets one (typically small) universal prior weight $2^{-K(\mu_{\text{general}})}$ in the sum defining $\Upsilon(\pi)$.

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From “Is AGI Possible?” to “When Will AGI Arrive?”

With the rise of increasingly general and powerful Large Language Models (LLMs), the conversation has shifted from *whether* AGI is possible to *when* it will arrive, and whether current systems already count as (proto-)AGI.

Key claim of this section: the language-modelling path to AGI is *not* in opposition to Universal Artificial Intelligence – it is arguably an **approximation** of the universal Bayesian mixture $\xi(x_i)$ that AIXI uses. LLMs are trained on huge corpora of text with the goal of minimizing **cross-entropy loss** (a measure of how surprised the model is by the next token). To minimize this loss efficiently, the model must build an internal representation of the knowledge in its training data – much like a data **compressor** does. As stated earlier in the book: *compression is prediction*, and Kolmogorov complexity / Solomonoff induction are the theoretical “ultimate” compressors/predictors.

LLMs, Planning, and MC-AIXI

LLMs alone are not complete reinforcement-learning/AI agents, but recent work augments them in ways that mirror this book's own extensions of Solomonoff induction:

- **Reasoning-via-Planning (RAP)** augments an LLM with planning methods, mirroring how **MC-AIXI** (Chapter 7 / 12) augments Solomonoff induction with Monte Carlo Tree Search (MCTS).
- **In-context learning** (an LLM's ability to adapt its behaviour from examples given in its prompt, without updating its weights) can be viewed as a form of **Bayesian inference** – and it has been shown that Transformers, when trained on suitably algorithmic data, can approximate Solomonoff induction directly.
- **RLHF** (Reinforcement Learning from Human Feedback) is a crude form of RL – essentially using a horizon of just 1 step, trained on a model of human preferences/values as a reward proxy.
- **Tree-of-Thought (ToT)**, a search/planning controller layered on top of an LLM, corresponds to **expectimax planning**/tree search; combined with in-context learning (the analogue of Solomonoff induction here), this yields something like an LLM version of MC-AIXI.

What Is Still Missing from LLMs, Relative to AIXI?

At the time of writing, LLMs are **not** AGI on their own. What would bring them closer to AIXI?

- **Long-horizon value functions.** RLHF trains on a horizon of essentially 1 step; learning a value function over much longer horizons would let models plan and reason at length, rather than react token-by-token.
- **New architectures, scaffolding, or controllers** around the base model (e.g. explicit search/planning modules, or tool use) to add the missing long-horizon planning capability.
- **Memory / continual learning.** Current LLMs largely lack a good mechanism to store, recall, and learn from past experience beyond their fixed context window. Possible directions: an explicit external memory controller, or continual learning that compiles short-term in-context experience into long-term (in-weight) memory – the latter would be closest in spirit to Solomonoff induction, which learns continuously and without a fixed context limit.

AIXI, by contrast, learns continuously, plans at arbitrarily long horizons, and handles arbitrary (multimodal) inputs/outputs by construction – serving as the gold-standard *direction* to aim for when designing the next generation of LLMs, towards AGSI.

Python: Compression Is Prediction

A concrete numeric illustration of “compression is prediction”: a *predictable* (structured) sequence compresses far better than a *random* one of the same length, exactly as an ideal Solomonoff-style predictor would exploit.

```
1 import zlib, math
2 import numpy as np
3 rng = np.random.default_rng(0)
4
5 n = 20000
6 structured = (bytes([65 + (i % 4) for i in range(n)]))      # "ABCDABCDABCD..."
7 random_bytes = bytes(rng.integers(0, 256, size=n, dtype=np.uint8))
8
9 for name, data in [("structured (period-4 pattern)", structured),
10                  ("random bytes", random_bytes)]:
11     compressed = zlib.compress(data, level=9)
12     ratio = len(compressed) / len(data)
13     bits_per_symbol = 8 * ratio
14     print(f"{name:32s}: raw={len(data)}B compressed={len(compressed)}B "
15           f"ratio={ratio:.4f} ~{bits_per_symbol:.3f} bits/symbol")
16 # structured -> tiny fraction of its size (~0.02 bits/symbol);
17 # random      -> barely compresses at all (~8 bits/symbol = true entropy).
```

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Towards Artificial General Super Intelligence (AGSI)

The construction of a truly intelligent system able to act beyond human capacity in most areas – **Artificial General Super Intelligence (AGSI)** – is one of the most ambitious problems of our time. If achieved, it represents the entering of a new era for humanity, and perhaps the last invention we ever need to make.

This book has provided the tools needed to understand and extend the **Universal Artificial Intelligence** approach to this problem – from its theoretical and safety/philosophical foundations, to the best currently implementable approximations (MC-AIXI, CTW, and beyond).

Chapter Summary

- **16.1** Universal induction rests on solid philosophical ground: Epicurus, Occam, and Bayes combine into a single formula, $w_\nu^U = 2^{-K(\nu)}$.
- **16.2–16.3** Consciousness, free will, and moral status remain open, but a pragmatic stance: if these properties are useful for acting intelligently, expect them to emerge in AIXI-like agents.
- **16.4** Whether AIXI-like agents self-replicate or teleport depends entirely on how their value function is defined (copy- vs. slot-centered).
- **16.5–16.6** Most informal arguments against AGI break down under formal scrutiny; arguments for AGI (PCT, Moore's Law, and especially concrete AI progress) are compelling and mounting.
- **16.7** The Legg–Hutter measure $\Upsilon(\pi)$ formalizes intelligence itself; AIXI, by construction, *maximizes* it.
- **16.8** LLMs are best understood as powerful *approximations* of Solomonoff induction, missing (for now) AIXI's long-horizon planning and continual memory.

Universal Artificial Intelligence: a rigorous theory of, and a formal path towards, AGI.